

Machine Learning Project Checklist (adapted from Aurelien Geron)

There are 8 main steps.

1. Get the data
2. Explore the data
3. Determine the question to investigate
4. Prepare the data
5. Model selection
6. Improving performance
7. Evaluate performance on the test set
8. Present your solution

Each of these steps can be broken down into smaller steps. When you make your presentation, make sure you can explain each step of the process. Depending on your specific problem, you may not need to do every step, but make sure to explain why you didn't need a particular step.

1. **Get the data**

- ☐ List the data you need
- ☐ Find and document where you can get the data
- ☐ Check how much space it the data takes up
- ☐ Check legal obligations and get authorization, if necessary
- ☐ Download the data or upload to your workspace (Colab)
- ☐ Convert the data into a format you can work with (without changing the data yet)
- ☐ Check if there is any private information that needs to be deleted or hidden
- ☐ Check the number of columns, column names, size of the dataset, and data types present

2. **Explore the data**

- ☐ Study each attribute and its characteristics
 - ☐ Feature names and what they mean
 - ☐ Data type
 - ☐ Missing values?
 - ☐ Outliers?
- ☐ Visualize the data
- ☐ Study correlations between attributes
- ☐ Identify if you might need to scale or transform the data
- ☐ Identify if there is any extra data that might be helpful
- ☐ Document what you have learned/discovered about the data

3. Determine the question to investigate

- ☐ Can it be answered with supervised or unsupervised learning?
 - ☐ Classification? Regression? Clustering? Dimensionality reduction? Something else?
- ☐ How will you determine if your model successfully answers your question or not?
- ☐ How would you solve the problem manually, without a computer?

4. Prepare the data

Note: try to avoid changing the original dataset. Instead, create new dataframes or make copies of the dataframe for the changes you make.

- ☐ Data Cleaning
 - ☐ Remove outliers (or not)
 - ☐ Fill in missing values or drop them (by dropping rows or dropping entire columns)
- ☐ Feature selection
 - ☐ Drop features which seem useless (or don't)
- ☐ Feature Engineering, where appropriate
 - ☐ One-hot encoding
 - ☐ Feature splitting (such as "total spending per week" into just "total spending")
 - ☐ Making new features (combining features, such as "z" and "n" into "z/n")
 - ☐ See if you need to transform features (log, square, square root, etc.)
- ☐ Scale or normalize the features
- ☐ Split the data into training and test sets (remember: never do anything on the test set until the very end when it's time to evaluate)

5. Model selection

- ☐ Select from classification or regression models, depending on your task
- ☐ Make an instance of the model and fit it to the training data.
 - ☐ Use default hyperparameters at first
 - ☐ Look up the model's documentation to see what some of them are
- ☐ Consider training multiple models and to measure and compare their performance
 - ☐ Helpful to use N-fold cross validation

6. Improving performance

- ☐ Use grid search or random grid search to optimize parameters for a given metric
- ☐ Try optimizing for different metrics, depending on which one matters most to you
- ☐ In the data, see if you can treat NaN values differently (just make them zero? Drop them? Median or mean imputation?)
- ☐ Remember NOT to use your test data yet...

7. Evaluate performance on test set

- ☐ Make another instance of the model you chose and give it the hyperparameters that you optimized in the previous step.
- ☐ Train the model on your training data
- ☐ Have your model make predictions on the X test data
- ☐ Compare the model's predictions `y_pred` with the true `y_test` data using appropriate metrics

8. Present your solution

- ☐ Document what you have done
- ☐ Create a nice presentation
 - ☐ Start broadly with the big picture first
 - ☐ Explain your thought process along the way
 - ☐ Use visuals and easy-to-remember statements for key points you want to emphasize
- ☐ Helpful to practice by writing a script
- ☐ Describe what worked and what didn't
- ☐ Don't forget to present any other points you find interesting
- ☐ Did your ML model successfully solve your problem/answer your question?

Bonus:

The code you wrote was for one machine learning project. Is there a way you could turn your code into individual functions which can be stored in a module? This way, you can more easily copy this same workflow for similar tasks.